THE SHEAF OF DIFFERENTIALS OF A MORPHISM OF ALGEBRAIC SPACES IS QUASI-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointer is located at its \label line inside the pinned chapter file; the two cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Local presentation of a morphism of algebraic spaces by a diagram of schemes. Same chapter, Lemma labelled lemma-localize-differentials (source line 981, Stacks tag 04CU); invoked at proof lines 1023-1024. Supplies the diagram with surjective a and b and U, V schemes, which is what reduces the statement to the scheme case.
- Quasi-coherence of the sheaf of differentials of a morphism of schemes. Morphisms, Lemma labelled morphisms-lemma-differentials-diagonal (canonical Stacks tag 08S2); invoked at proof line 1028. Supplies quasi-coherence of Omega_{U/V} in the local model.
- Criterion for quasi-coherence on an algebraic space by local presentation. Properties of Spaces, Lemma labelled spaces-properties-lemma-characterize-quasi-coherent (canonical Stacks tag 03M0); invoked at proof lines 1029-1031. Supplies the passage from quasi-coherence of the restriction to a presentation to quasi-coherence on X.

Source reference. Original lemma 04CU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–981. Same-chapter pointer invoked at proof line 1024 for the presentation diagram; the identification of the restricted differentials and the final quasi-coherence criterion are retained in the counted span.

Source reference. Original lemma 08S2: sources/stacks-chow/tags.txt, source lines 11115–11115. Cross-chapter standard prerequisite offered at proof line 1028 as one way to see quasi-coherence of Omega_{U/V}; the reduction through the presentation is retained in the counted span.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03M0: sources/stacks-chow/tags.txt, source lines 4732–4732. Cross-chapter standard prerequisite invoked once at proof lines 1029-1031 for the criterion that quasi-coherence on an algebraic space may be checked on a presentation.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Then $\Omega_{X/Y}$ is a quasi-coherent $\mathcal{O}_X$-module.*

Proof. Choose a diagram as in Lemma 04CU with a and b surjective and U and V schemes. Then we see that $\Omega_{X/Y}|_U = \Omega_{U/V}$ which is quasi-coherent (for example by Morphisms, Lemma 08S2). Hence we conclude that $\Omega_{X/Y}$ is quasi-coherent by Properties of Spaces, Lemma 03M0. □